GOOD NOTATION
THE DATA AND AI SCHOOL

DATA SCIENCE SYLLABUS

Beginner to Advanced | Industry-Oriented Learning

Python • SQL • Statistics • Machine Learning

Generative AI • LLMs • Agentic AI • MLOps

6-Month Learning Roadmap | Practical Labs | Portfolio Projects

Designed for students, freshers, working professionals and career-switchers.
Recommended guided learning: approximately 260 hours across six months, plus independent practice and project work.

Program Overview

A structured path from programming foundations to deployed machine learning and AI applications.

PROGRAM DETAIL	RECOMMENDATION
Duration	6 months
Learning style	Concepts, instructor-led demos, guided labs, assignments and projects
Prerequisites	Basic computer skills; no prior data science experience required
Core outcome	Analyze data, train and evaluate ML models, build GenAI prototypes, and deploy applications
Portfolio target	3–5 substantial projects with code, documentation and demonstrations

Six-Month Curriculum

MONTH 1	Python Programming and Statistics
BEGINNER	• Python setup, variables, data types, operators and input/output • Conditions, loops, functions, lambda and comprehensions • Lists, tuples, sets, dictionaries, OOP, exceptions and file handling • NumPy arrays, indexing, slicing, broadcasting and vectorization • Descriptive statistics, probability, distributions, variance and standard deviation • Correlation, sampling, confidence intervals, hypothesis testing and p-values **Practical:** Python business calculator and statistical analysis notebook.

MONTH 2	Data Analysis, SQL and Visualization
BEGINNER → INTERMEDIATE	• Pandas Series/DataFrames, filtering, grouping, merging and reshaping • Data cleaning: missing values, duplicates, outliers and data types • Exploratory Data Analysis (EDA) and data quality checks • Matplotlib, Seaborn and Plotly visualizations • SQL: SELECT, filtering, aggregation, joins, subqueries and CTEs • Window functions, views, indexes and stored procedures • Excel, PivotTables, Power Query, Power BI and introductory DAX **Practical:** Sales analysis, customer segmentation and interactive business dashboard.

MONTH 3	Machine Learning Fundamentals
INTERMEDIATE	• Scikit-learn workflow, preprocessing and train-test split • Cross-validation, pipelines and data leakage prevention • Linear/polynomial regression and logistic regression • Decision trees, random forests, KNN, Naive Bayes and SVM • Feature engineering, encoding, scaling and imbalanced data • Metrics: MAE, RMSE, R^2, precision, recall, F1-score and ROC-AUC • Overfitting, underfitting and hyperparameter tuning **Practical:** House-price prediction and customer-churn classification.

MONTH 4	Advanced Machine Learning
INTERMEDIATE → ADVANCED	• K-Means, hierarchical clustering and DBSCAN • PCA, dimensionality reduction and feature selection • Gradient boosting: XGBoost, LightGBM and CatBoost • Ensemble learning and systematic model comparison • Time-series trends, seasonality and forecasting • Recommendation systems and anomaly/fraud detection • Explainable AI: feature importance and SHAP • Reproducible experiments and model pipelines **Practical:** Customer segmentation, demand forecasting and fraud detection prototype.

MONTH 5	Deep Learning, Generative AI and LLMs
ADVANCED	• Neural networks, activation functions, loss and backpropagation • TensorFlow/Keras or PyTorch fundamentals • NLP, tokenization, embeddings and Transformer architecture • Large Language Models (LLMs), prompting and structured outputs • Calling AI model APIs from Python • Retrieval-Augmented Generation (RAG), chunking and retrieval • Vector databases, semantic search and evaluation • Fine-tuning concepts, hallucinations, privacy and responsible AI **Practical:** PDF/document question-answering chatbot with source references.

MONTH 6	Deployment, MLOps and Agentic AI
ADVANCED / CAPSTONE	• FastAPI and Streamlit applications • Git/GitHub, virtual environments and dependency management • Docker fundamentals and application deployment • Cloud fundamentals: choose AWS, Azure or Google Cloud • MLflow, experiment tracking, model versioning and reproducibility • CI/CD concepts, monitoring, data/model drift and retraining • Agentic AI: tools, planning, workflows and memory concepts • Agent testing, permissions, security, cost and latency optimization • Portfolio, resume, project walkthrough and mock interviews **Practical:** Deploy an ML prediction API and build an AI web-research agent.

Recommended Technology Stack

Teach one tool deeply in each category before introducing alternatives.

AREA	TOOLS / TECHNOLOGIES
Programming	Python, Jupyter Notebook, VS Code
Data analysis	NumPy, Pandas
Statistics	SciPy, Statsmodels
Visualization	Matplotlib, Seaborn, Plotly
Databases	SQL Server or MySQL
Machine learning	Scikit-learn, XGBoost; optionally LightGBM/CatBoost
Deep learning	PyTorch or TensorFlow/Keras
Generative AI	LLM APIs, embeddings, RAG frameworks and vector databases
BI / analytics	Excel, Power BI
Deployment	FastAPI, Streamlit, Docker
MLOps / cloud	Git, MLflow, CI/CD and one cloud platform

Capstone Project Portfolio

1. Retail Sales Intelligence

Technology: Python+Pandas+ SQL+ Power BI

Clean transactional data, calculate KPIs, analyze trends and present insights in a dashboard.

2. Customer Churn Prediction

Technology: Scikit-learn+feature engineering

Train and evaluate a classification model, explain important features and discuss business actions.

3. GenAI Document Assistant

Technology: Python+ LLM+RAG

Retrieve passages from uploaded documents and answer questions with source references.

4. AI Web Research Agent

Technology:LLM+ search/API tools+ agent workflow

Collect evidence from permitted sources, summarize findings and produce a structured report with citations.

5. End-to-End ML Deployment

Technology: Scikit-learn +FastAPI +Docker

Serve model predictions through an API and demonstrate versioning and basic monitoring.

Suggested Training Hours

MONTH	FOCUS	GUIDED HOURS
1	Python and statistics	40
2	Pandas, SQL and visualization	40
3	Machine learning	45
4	Advanced ML	45
5	Deep learning and GenAI	45
6	Deployment, MLOps and capstone	45
TOTAL	**Guided classes and practicals**	**260**

Independent practice and project work should be scheduled in addition to guided hours. Adjust the pace based on students' programming background and assessment results.

Assessment and Completion Checklist

- Weekly coding exercises and short quizzes.
- SQL practice using realistic relational datasets.
- Data cleaning and EDA assignment with a written insight summary.
- At least two ML assignments with appropriate evaluation metrics.
- One GenAI/RAG prototype and one deployed application.
- Final capstone presentation, GitHub repository and project documentation. •

Resume review, technical interview practice and mock project explanation.

CAREER-READINESS OUTCOME

Students should be able to explain the full workflow: define a business problem → collect and clean data → analyze and model → evaluate results → communicate insights → deploy responsibly.

Recommended Learning Resources

Microsoft Learn — Data Scientist career path
https://learn.microsoft.com/en-us/training/career-paths/data-scientist

Google — Machine Learning Crash Course
https://developers.google.com/machine-learning/crash-course

Pandas documentation
https://pandas.pydata.org/docs/

Scikit-learn user guide
https://scikit-learn.org/stable/user_guide.html

PyTorch tutorials
https://pytorch.org/tutorials/

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Curriculum note: This is a recommended 2026 learning plan. Specific libraries and cloud services may be updated as industry tools evolve.

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